

Section B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)	(i)		121 kJ mol ⁻¹		1		1		
		(ii)		$\Delta H = -(-728) - (242) - (1735) - (178) + (-795)$ (1) $\Delta H = -2222$ kJ mol ⁻¹ (1)		2		2	1	
		(iii)		award (1) for any value in range 900-1600 kJ mol ⁻¹ award (1) for either of following - ionisation energies increase for successive ionisation energies (so value must be more than half the ionisation of the first two electrons) - second ionisation energy is greater than first ionisation energy		1	1	2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		at minimum temperature $\Delta G = 0$ so equation gives $\Delta S = \Delta H/T$ (1) change units giving $T = 473 \text{ K}$ and 78000 J mol^{-1} (1) $\Delta S = 165 \text{ J K}^{-1} \text{ mol}^{-1}$ (allow 164.9) (1) ECF possible	1	1		3	3	
		(ii)		0.0284 mol of CaCl_2 (1) 0.1138 mol of water (1) $x = 4$ (1) ECF possible		3		3	2	
		(iii)		brick-red	1			1		1
	(c)			CaCl_2 misty fumes (and white solid) (1) CaBr_2 will produce orange fumes / solution as well as misty fumes (and white solid) (1) accept will also produce orange fumes if first mark awarded bromide is easier to oxidise than chloride (so can be oxidised by the sulfuric acid) (1)	3			3		1 1
				Question 9 total	5	9	1	15	6	3